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The impact of media coverage on the ESG performance of heavily polluting enterprises in China

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Abstract
Based on panel data from 2011 to 2021 encompassing 390 listed enterprises in China, we investigate the potential influence of media coverage on the Environmental, Social, and Governance (ESG) performance of heavy-polluting enterprises. The study discerns that positive media coverage significantly contributes to an enhanced ESG performance among heavy-polluting enterprises, particularly those in eastern and western China and those entrenched within highly competitive industries. Positive media coverage predominantly amplifies these enterprises' environmental and social performance dimensions, while intriguingly, negative media coverage unexpectedly positively impacts their environmental performance. Green total factor productivity and green innovation emerge as crucial facilitators in bolstering the relationship between positive media coverage and the ESG performance of these enterprises. Our research findings provide valuable insights into understanding enterprise ESG behavior and the role of media coverage.

Keywords Media coverage · ESG performance · Panel fixed effects

1 Introduction

With the development of global issues such as climate change at macro and micro levels, Environmental, Social, and Governance (ESG) performance has become a critical aspect of corporate responsibility and sustainable development [1, 2]. Investors and other stakeholders (including customers and regulatory bodies) are increasingly accustomed to utilizing ESG performance as a tool to scrutinize enterprises' environmental and social responsibility actions [3, 4]. The ESG disclosure of enterprises is no longer a passive activity but has evolved into a proactive choice for enterprises to showcase their image actively [5]. Robust ESG disclosure aids enterprises in maintaining reputation and brand value [6], reducing supply chain and compliance risks [7], and ensuring long-term stability. Media coverage has long held significant influence over enterprise operational activities [8]. As a crucial mechanism, enterprises can effectively publicize their initiatives and efforts in specific areas through the media [9]. Strategic use of media platforms allows enterprises to demonstrate their commitment to responsible and sustainable development practices, thereby enhancing transparency and accountability in alignment with ESG principles [10]. Additionally, media reporting provides a valuable channel for enterprises to respond to stakeholders' concerns proactively, facilitating constructive informed dialogues and the establishment of a positive enterprise reputation [11]. Consequently, it is imperative to ascertain the impact and mechanisms of media coverage on the ESG performance of enterprises, particularly those classified as heavy polluters.

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Scholarly investigations into the impact of media coverage on businesses have surged in recent years, reflecting the growing recognition of the media's influential role in shaping enterprise narratives. Research has focused on understanding the dynamics of media coverage, examining its role in shaping public opinion [12], influencing investor behavior [13], and affecting enterprise reputation [14]. With the advancement of digital technology, financial reporting is no longer confined to traditional mass media, and social media has emerged as an effective medium for stakeholders to access information [15]. This expansion of channels has elevated the influence of media coverage beyond any previous era. Research focusing on enterprise performance indicates that given investors' limited attention, media reporting significantly alters stock prices [16]. Negative coverage is detrimental to enterprises' market value, particularly for enterprises under intense media scrutiny, as their actions are magnified by the media, potentially causing more substantial losses in shareholder wealth [17]. The mechanisms driving these effects are complex, involving factors such as framing [18], crisis communication [19], and information asymmetry [20]. However, controversies persist in establishing causality [21], understanding media bias [22], evaluating the role of social media [23], and discerning between short-term and long-term effects [24].

The market has validated and confirmed the impact of ESG performance on market value. Particularly for heavily polluting enterprises characterized by industries with significant environmental footprints, there is mounting regulatory and societal pressure to address their environmental impact [25]. ESG performance, particularly in the environmental dimension, becomes paramount as heavy polluters navigate the imperative to minimize their ecological footprint, reduce emissions, and implement sustainable practices [26]. Adhering to stringent environmental standards not only ensures regulatory compliance but also contributes to mitigating adverse environmental effects and fostering long-term resilience [27]. The social dimension of ESG encompasses considerations related to employment, community engagement, and social responsibility [28]. For heavy polluters, whose operations often intersect with local communities, managing social impacts becomes crucial. Maintaining positive stakeholder relations, particularly with communities affected by industrial activities, is vital for securing a social license to operate [29]. ESG performance indicators related to social responsibility and community engagement become instrumental in shaping perceptions, garnering support, and mitigating potential conflicts [30]. The governance dimension of ESG underscores the importance of ethical business practices, transparency, and robust governance structures [31, 32]. For heavy polluters, governance integrity is essential not only for compliance with regulatory frameworks but also for fostering investor confidence [33]. ESG metrics related to governance, including board diversity, executive compensation transparency, and adherence to ethical standards, play a pivotal role in signaling corporate responsibility and accountability to investors, regulatory bodies, and the wider public [34, 35].

In previous studies, researchers have widely acknowledged that media scrutiny contributes to better corporate governance. However, it may also lead to a series of short-sighted behaviors by corporate decision-makers, who seek to increase the enterprise's exposure in the short term. There is no clear consensus among researchers on how critical or positive media coverage affects corporate ESG performance. Ge [36] pointed out that negative media coverage can effectively promote the improvement of corporate ESG performance, while Bissoondoyal-Bheenick et al. [37] argued that positive media coverage can reduce information asymmetry between enterprises and their investors regarding ESG investments. Moreover, media coverage is not always simply positive or negative; it can also be neutral. He et al. [38] found in a study of Chinese enterprises that regardless of the content, an increase in the overall volume of media coverage positively impacts corporate ESG performance. Conversely, good ESG performance can serve as a strong brand identity for an enterprise, which, when communicated through the media to investors and consumers, enhances the enterprise's competitiveness [39].

As the world's largest developing country, China's rapid industrialization and urbanization over the past few decades have brought environmental issues to the forefront, prompting the government to implement various policies to promote corporate sustainability. Additionally, the unique structure and regulatory environment of China's capital markets may mean that the impact of media coverage on corporate ESG performance differs from that in other countries. Therefore, investigating the relationship between media coverage and the ESG performance of heavily polluting enterprises in China can capture the complex interactions between the policy environment, economic development, and corporate behavior, providing empirical evidence for sustainable practices in developing countries.

We also recognize that not all enterprises are under the same level of environmental regulation pressure; heavily polluting enterprises are typically subject to the most public scrutiny and policy pressure [40]. The business activities of these enterprises directly impact environmental quality, making their ESG performance, particularly in environmental aspects, a frequent focus of media coverage. The media plays a crucial role not only in exposing pollution issues and monitoring policy implementation but also in applying public pressure to prompt enterprises to improve their practices. Furthermore, given strict environmental regulations and social responsibility requirements, these enterprises need to enhance their ESG performance to maintain and improve their reputation, with media coverage playing a key role in this process [41]. Thus, focusing on heavily polluting enterprises as the subject of research can reveal how the media influences public opinion and corporate reputation management, driving strategic adjustments in corporate environmental and social responsibility practices.

Under media scrutiny, existing research typically suggests that enterprises respond passively. For example, negative coverage can lead to greater financing constraints in the market, compelling enterprises to improve their ESG performance to alleviate financial pressure. However, in reality, enterprises can also adopt more proactive strategies in response to media coverage. They might choose to enhance their green innovation output and improve green innovation efficiency to demonstrate their commitment to sustainable practices to the public and investors. Therefore, we are curious whether media pressure can effectively motivate heavily polluting publicly listed enterprises in China to improve their ESG performance. Specifically, do these enterprises choose to proactively strengthen their green practices in response to media scrutiny, thereby gaining market favor and ultimately achieving better ESG outcomes? To address the questions outlined above, we conducted a study using panel data from 390 heavily polluting enterprises listed in China between 2011 and 2021. We applied a panel fixed-effects model to investigate whether media coverage in the Chinese market has improved the ESG performance of these enterprises and to examine the role of corporate green behavior, including green innovation and green total factor productivity, in this process. Additionally, we explored the heterogeneous effects of location, industry competition, and ESG sub-dimensions based on the above analysis. We further conducted robustness checks by altering variable measurements and model specifications to verify our baseline results. Our findings reveal that positive media coverage significantly improves the ESG performance of heavily polluting enterprises in China. The tests of moderating effects show that these enterprises strengthen the positive impact of media coverage on ESG performance by enhancing their green innovation and green total factor productivity. Furthermore, there is noticeable heterogeneity in the impact of media coverage on these enterprises. Specifically, positive media coverage has a stronger effect on improving the ESG performance of heavily polluting enterprises in the eastern and western regions compared to the central region. Enterprises in highly competitive industries experience a greater positive impact on their ESG performance from favorable media coverage. While both positive and negative media coverage significantly improve the environmental performance of heavily polluting enterprises, social performance is more notably driven by positive coverage, and governance performance appears not to be dependent on media coverage.

The potential contributions of our research are as follows. First, we conducted a comprehensive evaluation of how media coverage affects the ESG performance of heavily polluting enterprises in China by using panel fixed effects models and dynamic panel models, thereby significantly supplementing existing research. Second, we proposed a comprehensive theoretical framework that confirms the positive moderating role of corporate green innovation and green total factor productivity in the relationship between media coverage and ESG performance in heavily polluting enterprises. Our research demonstrates how companies can view media scrutiny as an opportunity and actively showcase their green practices to enhance ESG performance. Third, we expanded the discussion to examine the heterogeneity in the impact of positive and negative media coverage on ESG performance, as well as the potential effects of the enterprises' locations and industries. These analyses provide important policy references for corporate managers and policymakers to leverage media coverage and enhance its governance function to improve corporate ESG performance.

The remaining sections of this article are organized as follows. Section 2 presents the theoretical framework and hypotheses. Section 3 outlines the methodology employed. Section 4 showcases empirical results. The final section comprises the conclusion and policy implications.

2 Theoretical framework and hypotheses

2.1 Media coverage and ESG performance

According to media governance theory and information asymmetry theory, the media acts as an intermediary that transmits information to society, effectively reducing information asymmetry between enterprises and external stakeholders [42]. On one hand, media attention serves the function of conveying information. The higher the level of media attention an enterprise receives, the lower the degree of information asymmetry between the enterprise and external information users. When media attention increases, it enhances the disclosure of corporate information, subjects corporate behavior to external scrutiny, and increases the transparency of corporate information [43]. On the other hand, media attention has a supervisory function, as enterprises, based on reputation theory, commit to maintaining their reputation for future benefits. Media attention positively influences corporate managers' voluntary disclosure of environmental annual reports [44]. As an effective communication tool, media attention can showcase corporate governance achievements from environmental, social, and corporate perspectives, thereby influencing investors' decisions and reducing potential information asymmetry.

The power of media coverage as a potent external intervention force on enterprise ESG activities is widely discussed [38, 45]. Positive media coverage has been consistently linked to improved ESG performance among heavily polluting enterprises [46]. Scholars have shown that positive portrayals in the media can act as a catalyst for positive enterprise behavior, such as reducing carbon emissions [47, 48], decreasing pollutant emissions [49], and enhancing environmental offset investments [50]. Favorable reports often elevate enterprises' standing in the eyes of stakeholders, including investors, consumers, and regulators. It also triggers shareholder and employee recognition of enterprise actions [51]. This positive reinforcement can create a virtuous cycle where enterprises, motivated by positive media coverage, are incentivized to adopt and enhance sustainable practices [52]. Such practices, in turn, contribute to improved ESG performance metrics, reinforcing the positive image portrayed in the media [53].

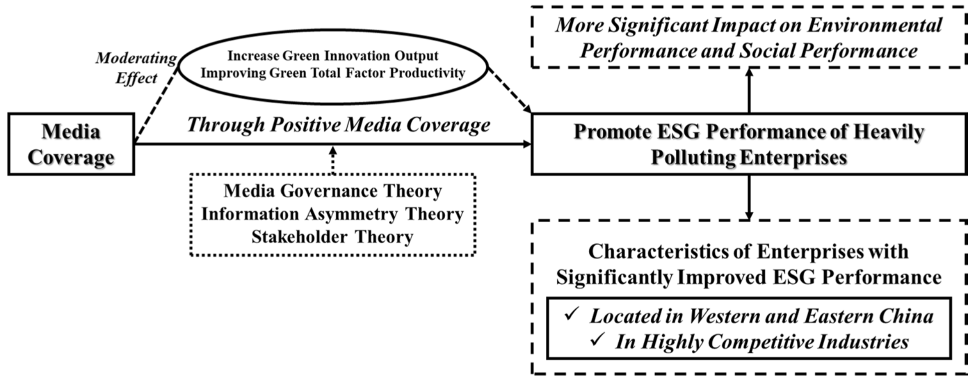
The influence of negative media coverage is contentious. Negative media reports expose deficiencies in environmental and social responsibility, leading to increased regulatory scrutiny and public pressure. Some scholars argue that negative news directly results in a decrease in market value, with no corresponding benefits from positive reporting [54]. Conversely, another group of researchers contends that the pressure from negative reporting swiftly translates to management and board levels, persuading enterprises to implement governance measures and instigate changes [55]. Negative signals enhance organizational innovation and learning capabilities, despite being a high-cost feedback for the enterprise, it is deemed necessary for improving enterprise behavior [56]. Heightened external attention often prompts enterprises to reassess and improve their environmental and social responsibility practices to mitigate reputation risks and unfavorable financial consequences [57]. In addition to linear relationships, some researchers suggest the existence of an inverted U-shaped relationship between negative media coverage and corporate social responsibility. It is posited that only when a certain level of negative media coverage is experienced, can corporate social responsibility compensate for the loss of legitimacy [58]. Thus, we propose hypotheses H1a and H1b.

H1a: Positive media coverage enhances the ESG performance of heavily polluting enterprises.
H1b: Negative media coverage enhances the ESG performance of heavily polluting enterprises.

2.2 Media coverage, green innovation and ESG performance

According to the Attention-Based View (ABV) theory, under pressure from public and social media scrutiny, governments and enterprises are likely to increase their focus on environmental issues, identifying them as opportunities rather than threats, and responding with appropriate adjustments [59]. For corporate managers, media coverage provides a window for enterprises to showcase their efforts in green practices to stakeholders. As a long-term strategy, green innovation activities require the media to communicate the progress and outcomes of these innovations to the public, thereby attracting more investors [60]. Green innovation aims to develop and implement new products, processes, services, or business models with positive environmental impacts, reducing ecological footprints and promoting sustainable development [61, 62]. Existing research confirms that green innovation initially leads to improved resource efficiency, including the use of renewable energy, reduced water consumption, and optimized material usage, with these technological advancements particularly significant in emerging market countries [63, 64]. The value of green innovation is also evident in the reduction of greenhouse gas emissions and other pollutants. Green improvements adopted in transportation, manufacturing, and energy production contribute to improving air and water quality, fostering sustainable environmental conditions [65–67]. The application of green innovation alleviates the environmental impact of traditional energy sources like fossil fuels, holding significant implications for promoting inclusive growth, creating employment opportunities, and enhancing public well-being [68, 69]. Due to the high input, high-risk, and low-return nature of innovative activities [70], external institutional factors become crucial catalysts for green innovation. Supportive regulatory frameworks and policies often influence and accelerate green innovation [71, 72]. Measures such as regulations, tax incentives, or subsidies to incentivize and mandate the adoption of green technologies have effectively propelled the growth of green innovation [73–75]. Scholars have also found that non-institutional factors impact green innovation activities, such as market preferences, community culture, and industry collaboration [76–78]. Media reporting is a unique external factor that imposes both institutional and non-institutional pressures on enterprises, conveying external attention in the form of institutional factors and transmitting non-institutional factors like community culture and market preferences to enterprises [79]. Media reporting makes enterprises constantly aware that they are under scrutiny and have

Fig. 1 Theoretical Framework



a responsibility to respond to stakeholder expectations [55]. Therefore, when sustainable development becomes a widely recognized issue, enterprises' willingness for green innovation is likely to increase. The application of green technologies and processes helps enterprises reduce emissions, improve resource efficiency, and achieve better environmental performance [80]. Green products and green processes create more job opportunities, demonstrating a sense of responsibility to the community [81]. Governance and structural changes aligned with green innovation contribute to enhancing overall governance levels and information transparency in enterprises [82]. Positive changes in these different dimensions are often confirmed to improve ESG performance [83, 84]. Therefore, we propose hypothesis H2.

H2: Green innovation enhances the positive impact of media coverage on ESG performance.

2.3 Media coverage, green total factor productivity, and ESG performance

Similar to green innovation, enterprises also have ample motivation to showcase improvements in green total factor productivity to stakeholders and other interested market participants through media coverage. Green TFP takes into consideration environmental factors and resource use efficiency alongside the traditional inputs and outputs. It aims to reflect not only economic performance but also the environmental sustainability of that performance [85]. There is limited research directly exploring the impact of media coverage on green total factor productivity. Based on the fundamental framework of green total factor productivity measurement, researchers have engaged in limited discussions. Media reports emphasize the financial benefits of green investments, such as long-term profit potential and resilience in the face of environmental challenges, attracting more green investments and socially responsible investments in green technologies and practices [86]. The influx of investments can stimulate innovation, contribute to the enhancement of green total factor productivity, and significantly improve ESG transparency [87]. Media coverage typically has a strong future orientation, and extensive reporting may foster a group of early adopters [88]. A typical example is that media reports reinforce the application of renewable energy and technologies, enhancing their diffusion levels [89]. Enterprises adjusting their energy structures and inputs can effectively improve their ESG performance [90]. Positive media coverage fosters knowledge sharing and flow within industries [91, 92], shapes consumer preferences [93], and enterprises may be more inclined to adopt and transform green technologies to achieve an enhancement in green total factor productivity. Thus, we posit hypothesis H3.

H3: Green total factor productivity enhances the positive impact of media coverage on ESG performance.

Figure 1 shows the overall research framework of this study.

3 Methodology

3.1 Empirical models

To investigate the overall impact of media coverage on the ESG performance of heavily polluting enterprises, we apply the following panel fixed-effects model.

$$ESG_{it} = c + \beta_1 Cover_{it} + \beta_2 Z_{it} + \lambda_i + \alpha_t + \varepsilon_{it} \tag{1}$$

where ESG_{it} represents the ESG performance score of enterprise i at time t. $Cover_{it}$ represents the media coverage of enterprise i during time t, which can be either positive or negative. Z_{it} represents a series of control variables. c denotes the constant term. λ_i is the individual fixed effect, α_t is the time fixed effect, and ε_{it} is the disturbance term.

Furthermore, to analyze the mechanisms through which media coverage influences the ESG performance of heavily polluting enterprises, we incorporate moderating effects into the baseline model. The model is specified as follows.

$$ESG_{it} = c + \beta_1 Cover_{it} + \beta_2 GI_{it} + \beta_3 Cover_{it} * GI_{it} + \beta_4 Z_{it} + \lambda_i + \alpha_t + \varepsilon_{it} \tag{2}$$

$$ESG_{it} = c + \beta_1 Cover_{it} + \beta_2 GTFP_{it} + \beta_3 Cover_{it} * GTFP_{it} + \beta_4 Z_{it} + \lambda_i + \alpha_t + \varepsilon_{it} \tag{3}$$

where GI_{it} represents the green innovation of enterprise i at time t. $GTFP_{it}$ represents the green total factor productivity of enterprise i at time t. $Cover_{it} * GI_{it}$ and $Cover_{it} * GTFP_{it}$ are the interaction terms between media coverage and green innovation, and media coverage and green total factor productivity, respectively. Other variables are the same as in Model (1).

In robustness tests, we applied a dynamic panel model and the Generalized Method of Moments (SYS-GMM Method) for re-estimation. Compared to the fixed-effects model, the dynamic panel model can effectively address endogeneity issues. We specify the model as follows.

$$ESG_{it} = c + \beta_1 L.ESG_{it} + \beta_2 Cover_{it} + \beta_2 Z_{it} + \lambda_i + \alpha_t + \varepsilon_{it} \tag{4}$$

where $L.ESG_{it}$ represents the lagged ESG performance of enterprise i at time $t - n$, i.e., the lagged term of the dependent variable.

3.2 Variable definitions and data sources

3.2.1 Dependent variable

The ESG performance of heavily polluting enterprises[1] serves as the dependent variable in this study. This data is sourced from Bloomberg's EGS ratings for Chinese publicly listed enterprises.[2] Bloomberg's ESG Index data dates back to 2009, ensuring the required time span for this study. Its indicator framework covers over 2,000 subfields under three primary categories, addressing various important topics related to sustainable development. Through data standardization, it effectively covers 80% of operational and employee conditions, thereby representatively reflecting the enterprise's operational impact. The Bloomberg ESG Index uses a bottom-up, model-driven approach, with information primarily derived from publicly disclosed enterprise data such as ESG reports, CSR reports, annual filings, proxy statements, corporate governance reports, and enterprise websites, aiming to facilitate a transparent and parameterized scoring system. It includes components for Environmental (E), Social (S), and Governance (G) aspects, with scores ranging from 0 to 100, where higher scores indicate better ESG performance by the enterprises.

3.2.2 Independent variables

Media coverage represents the level of attention that media gives to a certain type of event, typically measured by the total number of news articles related to that event. In this study, we use data from the financial news sub-database under

[1] According to the notice issued by the Ministry of Ecology and Environment of the People's Republic of China (formerly the Ministry of Environmental Protection of the People's Republic of China) titled "Notice on the Issuance of the <List of Industry Classifications for Environmental Protection Inspection of Listed Companies>," this study defines the following industries as heavily polluting industries: coal mining and washing industry, petroleum and natural gas extraction industry, black metal ore mining and dressing industry, non-ferrous metal ore mining and dressing industry, textile industry, leather, fur, feather, and its product and shoe-making industry, papermaking and paper products industry, petroleum processing, coking, and nuclear fuel processing industry, chemical raw materials and chemical product manufacturing industry, chemical fiber manufacturing industry, rubber and plastic products industry, non-metallic mineral products industry, black metal smelting and rolling processing industry, non-ferrous metal smelting and rolling processing industry, electricity, heat production, and supply industry.

[2] https://www.bloombergchina.com/solution/sustainable-finance/.

the Chinese Research Data Services Platform (CNRDS) as the source for media coverage. Additionally, using Python and the Chinese sentiment lexicon released by Dalian University of Technology,[3] we assess the overall sentiment of the news based on the sentiment of the words. Consequently, all news reports are categorized into positive, negative, and neutral reports. The specific classification rules are as follows: if the proportion of negative words exceeds 30%, the report is classified as negative; if the proportion of positive words exceeds 70%, it is classified as positive; the remaining reports are considered neutral. In the following research, we utilize data from positive and negative online news for the baseline empirical test. Additionally, we incorporate positive and negative newspaper coverage for robustness analysis. This arrangement takes into full consideration the role of the Internet in current market analysis. For investors and managers seeking to obtain more information with less effort, online financial news proves to be more efficient and influential compared to traditional media like newspapers [94].

3.2.3 Moderating variables

As we previously discussed, in response to media scrutiny, enterprises may enhance their public image by demonstrating better green practices. Green innovation, which focuses on technological improvements in areas such as negative environmental externalities, traditional energy consumption, and waste, makes the production process cleaner, reduces environmental pollution and resource consumption, and thus improves an enterprise's environmental performance [95]. On the other hand, green innovation enhances an enterprise's competitiveness in the market and generates more profits, which can also establish a good social reputation for the enterprise, thereby enhancing its social responsibility image [96]. Influenced by these factors, an enterprise's overall ESG performance is often improved. Further research has indicated that technology-standard environmental regulations promote corporate green total factor productivity through cost effects, innovation compensation effects, and factor allocation effects, although this relationship is not simply linear [97]. Therefore, we use green innovation and green total factor productivity to represent an enterprise's green practices. Green innovation is measured by the number of green patents filed by the enterprise, and the data is sourced from the CSMAR Green Patent Database.[4] The methodology for measuring green total factor productivity is adapted from the research process outlined by [98]. This data is sourced from Wind and disclosures in annual reports of listed enterprises.[5]

3.2.4 Control variables

Building on the research framework established by Yao et al. [99], this study incorporates various control variables, including firm size (*Size*), financial leverage ratio (*Lev*), Return on Assets (*ROA*), book-to-market ratio (*B/M*), Tobin's Q, liquidity (*Liquid*), cash flow (*Cashflow*), accounts receivable ratio (*REC*), proportion of independent directors (*Indep*), management shareholding ratio (*Mshare*), top ten shareholders' holding ratio (*TOP10*), firm age (*FirmAge*), and the duality of chairman and general manager roles (*Dual*). The data for the control variables are all sourced from the CSMAR database.

We then organize the aforementioned data into panel data by one-to-one matching based on listed enterprise codes and years. Then, using the "Notice on the Issuance of the < List of Industry Classifications for Environmental Protection Inspection of Listed Enterprises > " issued by the Ministry of Ecology and Environment of the People's Republic of China (formerly the Ministry of Environmental Protection of the People's Republic of China), the sample was filtered to include only heavily polluting enterprises. To address potential issues related to missing values and variations in regression results during empirical analysis, logarithmic transformations were applied to certain variables. To prevent the potential impact of outliers on the regression results, we further winsorized all continuous variables at 1% and 99%. The final sample consists of data from 384 listed heavily polluting enterprises in China, covering the period from 2011 to 2021, with a total of 3037 observations. Table 1 shows the symbols and definitions of all variables in this research.

[3] The detailed description and access link for the Chinese sentiment lexicon released by Dalian University of Technology can be found at: https://gitee.com/beyond277/DLUT-Emotionontology.

[4] https://data.csmar.com/.

[5] https://www.wind.com.cn/portal/en/EDB/index.html.

Table 1 Variable symbols and definitions

Symbols	Definitions
ESG	Measured by the ESG ratings of Chinese listed enterprises published by Bloomberg
PCover	If the proportion of positive words exceeds 70%, it is classified as positive
NCover	If the proportion of negative words exceeds 30%, the report is classified as negative
Size	The natural logarithm of an enterprise's total assets
Lev	The ratio of total liabilities to total assets
ROA	The ratio of net profit to total assets
BM	The ratio of book value to market value
TobinQ	The Tobin's Q value of the enterprise
Liquid	The ratio of current assets to current liabilities
Cashflow	The ratio of net cash flow to current liabilities
REC	The ratio of net accounts receivable to total assets
Indep	The ratio of independent directors to the total number of directors
Mshare	The ratio of management's shareholding to total share capital
TOP10	The shareholding ratio of the top ten shareholders
FirmAge	The current year minus the year of incorporation
Dual	Whether the chairman and the general manager are the same person

Table 2 Descriptive statistics

Variable	N	Mean	p50	SD	Min	Max
ESG	3037	3.313	3.330	0.327	2.161	4.228
PCover	3037	4.678	4.682	1.008	0.693	8.854
NCover	3037	4.263	4.277	1.066	0.000	8.925
Size	3037	3.298	2.996	1.949	0.000	12.68
Lev	3037	2.468	2.197	1.735	0.000	10.61
ROA	3037	23.35	23.34	1.422	19.20	28.64
BM	3037	0.507	0.518	0.201	0.014	2.290
TobinQ	3037	0.042	0.035	0.082	-1.125	0.929
Liquid	3037	0.744	0.764	0.264	0.039	1.430
Cashflow	3037	1.675	1.308	1.247	0.699	25.51
REC	3037	1.598	1.021	2.624	0.094	68.97
Indep	3037	0.069	0.066	0.079	− 1.686	0.664
Mshare	3037	0.058	0.039	0.061	0.000	0.596
TOP10	3037	37.04	33.33	5.395	23.08	66.67
FirmAge	3037	4.874	0.003	12.58	0.000	75.82
Dual	3037	60.93	61.02	16.70	12.71	98.59

4 Empirical results

4.1 Descriptive statistics

Table 2 presents the descriptive statistics for the variables in this study. We observe that the mean of ESG performance for heavily polluting enterprises in the sample is 3.313, which is lower than the median of 3.330. This suggests that more than half of the enterprises exhibit ESG performance superior to the average level of ESG performance across all sample enterprises. The mean values for positive and negative media coverage are 4.678 and 4.263, respectively, both lower than their respective medians of 4.682 and 4.277. This indicates that the positive and negative media coverage received by a majority of the enterprises falls below the average level of media coverage for the sample. Descriptive statistics for the various control variables reveal significant variations among them across different sample enterprises, and the implications of these differences will be further explored through in-depth empirical analysis.

Table 3 Benchmark results

	(1)	(2)	(3)	(4)
	ESG	*ESG*	*ESG*	*ESG*
PCover	0.015***		0.015**	
	(2.693)		(2.156)	
NCover		0.000		0.000
		(0.072)		(0.059)
Size	0.037***	0.043***	0.037**	0.043***
	(4.144)	(4.820)	(2.524)	(2.981)
Lev	– 0.139***	– 0.139***	– 0.139***	– 0.139***
	(– 4.104)	(– 4.120)	(– 3.028)	(– 3.041)
ROA	– 0.102*	– 0.088	– 0.102	– 0.088
	(– 1.887)	(– 1.634)	(– 1.634)	(– 1.438)
BM	– 0.073**	– 0.095***	– 0.073*	– 0.095**
	(– 2.490)	(– 3.306)	(-1.958)	(– 2.563)
TobinQ	– 0.006	– 0.006	– 0.006	– 0.006
	(– 1.299)	(– 1.303)	(– 1.070)	(– 1.066)
Liquid	– 0.004***	– 0.004***	– 0.004	– 0.004
	(– 2.974)	(– 2.950)	(– 1.347)	(– 1.329)
Cashflow	0.037	0.044	0.037	0.044
	(0.658)	(0.799)	(0.539)	(0.649)
REC	– 0.065	– 0.075	– 0.065	– 0.075
	(– 0.631)	(– 0.730)	(– 0.487)	(– 0.558)
Indep	0.002**	0.002**	0.002	0.002
	(2.249)	(2.327)	(1.573)	(1.610)
Mshare	– 0.000	– 0.000	– 0.000	– 0.000
	(– 0.580)	(– 0.452)	(– 0.528)	(– 0.407)
TOP10	0.001	0.000	0.001	0.000
	(1.209)	(1.074)	(0.742)	(0.660)
FirmAge	0.105	0.101	0.105	0.101
	(1.567)	(1.499)	(0.941)	(0.888)
Dual	– 0.004	– 0.003	– 0.004	– 0.003
	(– 0.360)	(– 0.280)	(– 0.268)	(– 0.207)
Constants	1.815***	1.783***	1.815***	1.783***
	(7.540)	(7.400)	(4.400)	(4.303)
Cluster	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
N	2809	2809	2809	2809
R^2	0.729	0.728	0.729	0.728

t statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

4.2 Benchmark regression results

Firstly, we conducted collinearity tests to mitigate potential collinearity risks. The mean Variance Inflation Factor (VIF) was reported as 1.74, confirming the absence of significant collinearity among the selected variables. Subsequently, we present the results of the baseline regression in Table 3. Results from columns (1) and (2) indicate that positive media coverage contributes to the improvement of ESG performance for heavily polluting enterprises, with a coefficient of 0.015, significant at the 1% level. However, the coefficient for the variable *NCover* is not significant, suggesting that heavily polluting enterprises do not exhibit a distinct response to negative media coverage. The same results are obtained after clustering, as shown in columns (3) and (4), where the coefficient for *PCover* is significant at the 5% level, while the variable *NCover* remains insignificant. Media coverage serves as a vital communication channel between stakeholders and enterprises. Enterprises with more positive media coverage can effectively reduce information asymmetry among stakeholders, allowing them to showcase their ESG performance and enhance attractiveness in this agenda [37]. Negative media coverage is undoubtedly undesirable for enterprises, and some may attempt to offset unfavorable reports by providing positive information through other channels [100]. However, such targeted masking behaviors do not lead to improvements in ESG performance. We also observed the impact of other variables on ESG performance. There is a positive correlation between firm size (*Size*) and ESG performance. Larger enterprises, benefiting from stronger capabilities, are more likely to achieve better ESG performance. Both financial leverage ratio (*Lev*) and book-to-market ratio (*B/M*) exhibit negative correlations with ESG performance. High financial leverage and B/M ratios imply more debt and risk, prompting experienced managers to control them within reasonable limits to demonstrate ambition to ESG investors while avoiding unnecessary debt pressure [101, 102]. In the absence of individual clustering analysis, liquidity (*Liquid*) is negatively correlated with ESG performance. Excessive liquidity may indicate poor profitability of enterprise assets to ESG stakeholders. However, this is not absolute, as researchers generally believe that good stock liquidity is an indication of a good reputation, promoting higher ESG ratings [103]. The proportion of independent directors (*Indep*) is positively correlated with ESG performance, consistent with existing research, suggesting that increasing the proportion of independent directors and other strategies to enhance board diversity can effectively improve the ESG image [34, 104].

4.3 Heterogeneity analysis

Next, we conducted a series of heterogeneity analyses to clearly illustrate the differences in media coverage of ESG performance. Enterprises' operation is influenced by locational factors, and cultural, policy, and ecological factors between geographic locations potentially influence media coverage of enterprises. Likewise, enterprise culture and managerial behavior patterns are shaped by these factors [105, 106]. Therefore, we grouped enterprises based on their registered locations in the eastern, central, and western regions of China and conducted regression analyses. The results are presented in Table 4. We observed that negative coverage still did not have an impact on all enterprises, regardless of their geographic location. Positive coverage had a significant positive effect on the ESG performance of heavily polluting enterprises in the eastern and western regions. This indicates that, compared to the central region, heavily polluting enterprises in the eastern and western regions are more likely to strive to improve their ESG performance in response to incentives from media coverage. This variation may stem from the higher economic and technological levels in the East, more evident policy support in the West, and pressure from central environmental protection inspections [107, 108].

Table 4 Heterogeneity analysis—regional

	East		Mid		West	
	(1)	(2)	(3)	(4)	(5)	(6)
	ESG	*ESG*	*ESG*	*ESG*	*ESG*	*ESG*
PCover	0.017**		0.003		0.028**	
	(2.143)		(0.266)		(2.047)	
NCover		− 0.004		0.010		− 0.009
		(− 0.506)		(0.973)		(− 0.750)
Constants	2.425***	2.371***	0.397	0.436	1.405**	1.276**
	(7.041)	(6.873)	(0.614)	(0.676)	(2.489)	(2.253)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1477	1477	769	769	513	513
R^2	0.741	0.741	0.712	0.712	0.760	0.758

t statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

Table 5 Heterogeneity analysis—industry competitiveness

	High industry competition		Low industry competition	
	(1)	(2)	(3)	(4)
PCover	0.018**		0.006	
	(2.227)		(0.781)	
NCover		0.001		0.004
		(0.114)		(0.574)
Constants	2.090***	2.068***	1.488***	1.473***
	(4.865)	(4.800)	(4.047)	(4.002)
Control	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
N	1280	1281	1529	1528
R^2	0.699	0.698	0.735	0.735

t statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

Table 6 Heterogeneity analysis—ESG sub-indicators

	Environmental		Social		Governance	
	(1)	(2)	(3)	(4)	(5)	(6)
PCover	0.152***		0.020*		− 0.003	
	(4.271)		(1.939)		(− 0.572)	
NCover		0.085**		− 0.004		− 0.005
		(2.527)		(− 0.393)		(− 1.205)
Constants	− 3.470**	− 3.765**	− 0.106	− 0.148	4.370***	4.367***
	(− 2.117)	(− 2.294)	(− 0.232)	(− 0.323)	(20.987)	(20.990)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2427	2427	2743	2743	2799	2799
R^2	0.421	0.417	0.285	0.283	0.703	0.703

t statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

The popularity of ESG performance has made enterprise initiatives in environmental protection crucial. Faced with pressure from the media, enterprises in different industries may adopt varying responses based on the intensity of competition within their industry. In existing research, the Herfindahl index is often applied to measure industry competition [109]. This index is a (0,1) measure, with values closer to 1 indicating a more concentrated market, a higher likelihood of monopoly, and weaker market competitiveness. We categorized enterprises with an index below the average as the high-competition industry group and those with an index above the average as the low-competition industry group. Table 5 presents the results of the heterogeneity analysis based on industry competition. Columns (1) and (2) display results for the high-competition industry group, while columns (3) and (4) show results for the low-competition industry group. Media coverage did not stimulate the low-competition industry group. However, positive coverage had a significant positive impact on the high-competition industry group. Faced with intense competitive environments and strong competitors, enterprises have ample motivation to enhance their ESG performance to gain market positions and maintain a competitive advantage [110].

In addition to enterprise characteristics, we were also curious about how enterprises respond to media coverage across the three dimensions of environment, social, and governance (ESG). Therefore, we further employed sub-indicators of ESG performance provided by Bloomberg for empirical regression, and the results are presented in Table 6. Firstly, positive media coverage had a positive impact on the environmental and social performance of enterprises, while its effect on governance performance was not significant. Existing research has found that positive media portrayal of an enterprise's social responsibility activities can help it stand out in fierce competition. However, when the media infers the business motives from the enterprise's social performance, this positive relationship may no longer exist [111]. Secondly, we observed a positive correlation between negative coverage and environmental performance. The rise of ESG scoring is closely related to sustainable development concepts, and environmental performance is often a focal point for both the media and stakeholders. Particularly for heavily polluting enterprises, negative coverage will stimulate them to improve their environmental performance, enhancing the frequency and quality of environmental information disclosure [112, 113].

4.4 The role of green innovation and green total factor productivity

A plethora of studies indicate that heavily polluting enterprises, when facing demands for sustainability and ESG performance pressures, need to achieve goals such as pollution reduction and emission reduction through means like technological enhancement and efficiency improvement [114]. Therefore, we analyzed the moderating effects brought by green innovation (GI) and green total factor productivity (GTFP), with results presented in Table 7. Columns (1) and (2) represent the impact of green innovation, while columns (3) and (4) represent the impact of green total factor productivity. We focused on the regression results of the interaction terms. The variable *PCover*GI* is significantly positive, indicating that

Table 7 Moderating effects analysis

	(1) ESG	(2) ESG	(3) ESG	(4) ESG
Panel A: Moderating Effects—Green Innovation				
PCover	0.011*			
	(1.942)			
NCover		− 0.002		
		(− 0.303)		
GI	− 0.032	− 0.010		
	(− 1.488)	(− 0.546)		
*PCover*GI*	0.009**			
	(1.991)			
*NCover*GI*		0.004		
		(1.143)		
Panel B: Moderating Effect—Green Total Factor Productivity				
PCover			− 0.073**	
			(− 2.029)	
NCover				− 0.097***
				(− 2.813)
GTFP			− 0.615**	− 0.608**
			(− 2.404)	(− 2.509)
PCover GTFP*			0.089**	
			(2.488)	
NCover GTFP*				0.098***
				(2.874)
Constants	1.824***	1.793***	2.532***	2.442***
	(7.584)	(7.438)	(7.305)	(7.486)
Control	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
N	2809	2809	2800	2800
R^2	0.730	0.728	0.730	0.730

t statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

positive media coverage increases the green innovation output, thereby helping the enterprise continuously improve its ESG performance. The same effect is reflected in green total factor productivity, where the variable *PCover* GTFP* is also significantly positive. Additionally, negative coverage can stimulate enterprises to enhance green total factor productivity and consequently improve their ESG performance. Our findings align with existing research. The willingness of enterprises to choose green innovation solutions depends on their awareness of social and environmental issues, and media coverage influences managerial behavior, contributing to the enhancement of this awareness. If enterprises prioritize social welfare and environmental protection and embrace social responsibility, they are more likely to choose strategies that involve developing green innovation and increasing green total factor productivity [115, 116].

4.5 Robustness tests

In comparison to online news, newspaper coverage continues to have a significant impact on enterprises. We employ newspaper reports as an explanatory variable for robustness checks. Additionally, considering potential endogeneity and the lagged effects of media coverage, we utilize a dynamic panel model to provide further robustness check results. These results are presented in Table 8. Columns (1) and (2) list the regression results when using newspaper reports as the explanatory variable. Similar to the benchmark regression results, the coefficient for the variable *PCover* is 0.008 and significant at the 10% level, suggesting that positive coverage from newspapers also enhances the ESG performance of heavily polluting enterprises. Columns (3) and (4) display the regression results using the dynamic panel model. We observe that the AR1 test results indicate the necessity of including lagged dependent variables in the equation. The AR2 test results show no autocorrelation in the disturbance terms. The p-value of the Sargan test is greater than 0.1, indicating the reliability of the instrumental variables used in the estimation. With a reasonable model specification confirmed, the variable *PCover* remains significantly positive at the 1% level, further supporting our benchmark regression results.

After conducting robustness checks on the baseline regression results, we proceeded to test the robustness of the moderating effects of green innovation and green total factor productivity. In this examination, we applied two strategies.

Table 8 Robustness test of benchmark regression

	(1) ESG	(2) ESG	(3) ESG	(4) ESG
Panel A: Replacement of Explanatory Variables (Newspapers)				
PCover (Newspaper)	0.008*			
	(1.667)			
NCover (Newspaper)		– 0.001		
		(– 0.195)		
Panel B: Changing Empirical Model (SYS-GMM)				
L.ESG			0.504***	0.503***
			(19.335)	(19.144)
PCover			0.017***	
			(2.865)	
NCover				– 0.003
				(– 0.469)
Constants	1.802***	1.805***	– 1.321***	– 1.307***
	(7.321)	(6.916)	(– 5.319)	(– 5.192)
AR(1) (P-value)	–	–	0.000	0.000
AR(2) (P-value)	–	–	0.417	0.323
Sargan (P-value)	–	–	0.972	1.000
Control	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	–
Individual FE	Yes	Yes	–	–
N	2751	2557	2512	2512
R^2	0.729	0.723	–	–

t statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

First, we treated news coverage as the dependent variable. Second, we utilized a panel fixed effects model with Driscoll-Kraay standard errors, known as the SCC model. This model fully accounts for heteroskedasticity and serial correlation issues, making the estimation results more robust [117]. Table 9 presents the results of this robustness check. The results in Panel A, Column (1) show that the regression coefficient of the variable *PCover*GI* is 0.001, which is significant at the 10% level. The results in Columns (3) and (4) indicate that the regression coefficients for the variables *PCover*GTFP* and

Table 9 Robustness test of the moderating effect

	(1) ESG	(2) ESG	(3) ESG	(4) ESG
Panel A: Replacement of explanatory variables (Newspapers)				
PCover	0.008*		− 0.063**	
	(1.647)		(− 2.255)	
NCover		− 0.001		− 0.069**
		(− 0.181)		(− 2.166)
GI	0.011	0.013		
	(1.348)	(1.556)		
*PCover*GI*	0.001*			
	(0.442)			
*NCover*GI*		− 0.001		
		(− 0.320)		
GTFP			− 0.463**	− 0.411*
			(− 2.096)	(− 1.813)
PCover GTFP*			0.070**	
			(2.571)	
NCover GTFP*				0.068**
				(2.166)
Constants	1.802***	1.805***	2.318***	2.258***
	(7.321)	(6.917)	(7.382)	(6.892)
Panel B: Changing Empirical Model (Panel SSC Methodl)				
PCover	0.011**		− 0.073*	
	(2.469)		(− 2.223)	
NCover		− 0.010		− 0.097***
		(− 0.602)		(− 3.387)
GI	− 0.032*	− 0.002		
	(− 2.047)	(− 0.838)		
*PCover*GI*	0.009**			
	(2.807)			
*NCover*GI*		0.004		
		(1.197)		
GTFP			− 0.615**	− 0.608***
			(− 3.119)	(− 3.713)
PCover GTFP*			0.089**	
			(2.696)	
NCover GTFP*				0.098***
				(3.297)
Constants	1.824***	1.793***	2.532***	2.442***
	(6.858)	(6.816)	(8.917)	(9.087)
Control	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
N	2809	2809	2800	2800
R^2	0.730	0.728	0.730	0.730

t statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

*NCover*GTFP* are 0.070 and 0.068, respectively, both significant at the 5% level. These results suggest that promoting green innovation and enhancing green total factor productivity are crucial channels through which positive media coverage improves an enterprise's ESG performance. Our previous analysis is thus credible.

5 Conclusions and policy implications

The pressure from the media typically holds significant sway over enterprises' ESG activities. We utilized panel data from 390 heavily polluting listed enterprises in China between 2011 and 2021 to investigate the impact of media coverage on ESG performance. Given the diversity in enterprise characteristics, we further examined variations in ESG performance exhibited by enterprises in different regions and industries in response to media coverage. We also took into account the potentially distinct influences on environmental, social, and governance performances. To enhance the robustness of our conclusions, we employed dynamic panel models and replaced explanatory variables. Furthermore, we attempted to analyze the roles of green innovation and green total factor productivity in the relationship between media coverage and ESG performance. Empirical results indicate that positive media coverage effectively enhances ESG performance, especially for those located in the eastern and western regions and those operating in highly competitive industries. The influence of positive media coverage on environmental and social performances continues to grow, while negative coverage contributes to enterprises improving their environmental performance. Mechanism analysis reveals that green innovation and the growth of green total factor productivity strengthen the positive impact of positive media coverage on ESG performance.

Our research findings hold significant practical value for both policymakers and corporate leaders. First, the empirical evidence highlights a noteworthy trajectory wherein positive media coverage not only correlates with an augmentation of environmental and social performance but also acts as a catalyst for continual growth in these dimensions. This signifies a sustained positive impact of media in steering heavy polluters toward higher standards of environmental conservation and social responsibility. Equally intriguing is the revelation that negative media coverage, conventionally perceived as detrimental, paradoxically stimulates enterprises to improve their environmental performance. This finding underscores the dual role of media, acting as both a source of commendation and a mechanism for constructive critique, thereby contributing to the overall enhancement of enterprises' environmental endeavors. As media coverage can effectively enhance ESG performance, especially through positive reporting, enterprises should consider proactive engagement with the media as part of their overall ESG strategy. By promoting their green innovation efforts and achievements in sustainability, enterprises can not only improve their public image but also potentially gain competitive advantages in the market.

Second, policymakers and enterprise decision-makers should advocate for the application and development of green technologies in a broader context and leverage media as a tool for promoting environmental and social governance. Green innovation emerges as a linchpin, wherein positive media coverage serves as an impetus for enterprises to engage in environmentally friendly innovation, thereby fostering a symbiotic relationship between media portrayal and technological advancement in sustainability. Similarly, the positive nexus between media coverage and green total factor growth rates accentuates the significance of holistic productivity in amplifying the positive influence of media on enterprises' overall ESG performance. These findings collectively emphasize the interconnectedness of media dynamics, innovation, and productivity in shaping the sustainability landscape of China's heavy polluters.

Finally, the policy implications derived from these empirical revelations underscore the need for a targeted and nuanced approach to media engagement and coverage within the regulatory framework. Policymakers ought to recognize the region-specific and industry-sensitive nature of media influence on heavy polluters, tailoring interventions that consider these nuances. Encouragingly, the findings also suggest that both positive and negative media coverage can act as catalysts for environmental improvements, signaling an opportunity for policymakers to leverage media dynamics as a mechanism for steering heavy polluters toward more sustainable practices. Additionally, fostering an environment that nurtures green innovation and holistic productivity is paramount, as these factors emerge as pivotal drivers that magnify the positive impact of media coverage on enterprises' ESG performance. As the policy landscape continues to evolve, a nuanced understanding of the symbiotic relationship between media, innovation, and productivity is crucial for the formulation of effective and targeted strategies aimed at promoting sustainable practices among heavy polluters in China.

Despite our research exploring the relationship between media coverage, corporate green practices, and the ESG performance of heavily polluting enterprises, there are inevitable limitations and shortcomings due to the author's limited research capabilities and the challenges in obtaining relevant data. We believe that further investigations will reveal more interesting findings on how media coverage reshapes and influences corporate ESG practices. In future research, we plan to incorporate text analysis from social media into the media coverage variable, which will help us examine the relationship between media and corporate ESG practices from a broader perspective. Moreover, in China's unique market environment, the mechanisms by which media coverage affects corporate ESG performance deserve further exploration. These influences are not limited to heavily polluting enterprises; factors such as environmental regulation, market oversight, and stock prices are also closely related.

Author contributions Sen Li: Thesis Design, Methodology, Data Processing, Draft Writing, Revision Han Long: Draft Writing, Revision.

Data availability The main data sources for this study are as follows: https://www.bloombergchina.com/solution/sustainable-finance/. https://data.csmar.com/. https://www.wind.com.cn/portal/en/EDB/index.html. The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

References

1. Broadstock DC, Chan K, Cheng LTW, Wang X. The role of ESG performance during times of financial crisis: evidence from COVID-19 in China. Financ Res Lett. 2021;38: 101716. https://doi.org/10.1016/j.frl.2020.101716.
2. Feng G, Long H, Wang H, Chang C. Environmental, social and governance, corporate social responsibility, and stock returns: What are the short- and long-Run relationships? Corp Soc Responsib Environ Manage. 2022. https://doi.org/10.1002/csr.2334.
3. Gillan SL, Koch A, Starks LT. Firms and social responsibility: a review of ESG and CSR research in corporate finance. J Corp Finan. 2021;66: 101889. https://doi.org/10.1016/j.jcorpfin.2021.101889.
4. Moreira OJ, Rodrigues MCM. Sourcing third party logistics service providers based on environmental, social and corporate governance: a case study. Discov Sustain. 2023;4(1):36. https://doi.org/10.1007/s43621-023-00149-3.
5. Xie J, Nozawa W, Yagi M, Fujii H, Managi S. Do environmental, social, and governance activities improve corporate financial performance? Bus Strateg Environ. 2019;28(2):286–300. https://doi.org/10.1002/bse.2224.
6. Ma D, Lv B, Liu Y, Liu S, Li X. Brand premium and carbon information disclosure strategy: evidence from China listed companies. Sustainability. 2023;15(6):5240. https://doi.org/10.3390/su15065240.
7. Sardanelli D, Bittucci L, Mirone F, Marzioni S. An integrative framework for supply chain rating: from financial-based to ESG-based rating models. Total Qual Manag Bus Excellence. 2022. https://doi.org/10.1080/14783363.2022.2069557.
8. Dyck A, Volchkova N, Zingales L. The corporate governance role of the media: evidence from Russia. J Financ. 2008;63(3):1093–135. https://doi.org/10.1111/j.1540-6261.2008.01353.x.
9. Ahern KR, Sosyura D. Who Writes the News? Corporate press releases during merger negotiations. J Financ. 2014;69(1):241–91. https://doi.org/10.1111/jofi.12109.
10. Kölbel JF, Busch T, Jancso LM. How media coverage of corporate social irresponsibility increases financial risk: media coverage of corporate social irresponsibility. Strateg Manag J. 2017;38(11):2266–84. https://doi.org/10.1002/smj.2647.
11. Solomon DH. Selective publicity and stock prices. J Financ. 2012;67(2):599–638. https://doi.org/10.1111/j.1540-6261.2012.01726.x.
12. Neresini F, Lorenzet A. Can media monitoring be a proxy for public opinion about technoscientific controversies? The case of the Italian public debate on nuclear power. Public Underst Sci. 2016;25(2):171–85. https://doi.org/10.1177/0963662514551506.
13. Wu C-H, Lin C-J. The impact of media coverage on investor trading behavior and stock returns. Pac Basin Financ J. 2017;43:151–72. https://doi.org/10.1016/j.pacfin.2017.04.001.
14. Wong JB, Zhang Q. Stock market reactions to adverse ESG disclosure via media channels. Br Account Rev. 2022;54(1): 101045. https://doi.org/10.1016/j.bar.2021.101045.
15. Dong H, Ren J, Padmanabhan B, Nickerson JV. How are social and mass media different in relation to the stock market? A study on topic coverage and predictive value. Inform Manage. 2022;59(2): 103588. https://doi.org/10.1016/j.im.2021.103588.
16. Liu LX, Sherman AE, Zhang Y. The long-run role of the media: evidence from initial public offerings. Manage Sci. 2014;60(8):1945–64. https://doi.org/10.1287/mnsc.2013.1851.

17. Xu XD, Zeng SX, Zou HL, Shi JJ. The impact of corporate environmental violation on shareholders' wealth: a perspective taken from media coverage. Bus Strateg Environ. 2016;25(2):73–91. https://doi.org/10.1002/bse.1858.
18. Colistra R, Johnson CB. Framing the legalization of marriage for same-sex couples: an examination of news coverage surrounding the U.S. supreme court's landmark decision. J Homosexuality. 2021;68(1):88–111. https://doi.org/10.1080/00918369.2019.1627128.
19. Nijkrake J, Gosselt JF, Gutteling JM. Competing frames and tone in corporate communication versus media coverage during a crisis. Public Relations Rev. 2015;41(1):80–8. https://doi.org/10.1016/j.pubrev.2014.10.010.
20. Chen X, Gong X, Yang Z. Media report favoritism and consequences: a comparison of traditional and new energy sector. Energy Econ. 2021;104: 105657. https://doi.org/10.1016/j.eneco.2021.105657.
21. Khalifa M, Sheikhbahaei A, Sualihu MA. The power of the business media: evidence from firm-level productivity. J Bus Finance Account. 2023. https://doi.org/10.1111/jbfa.12698.
22. Oliver AG, Campbell R, Graffin S, Bundy J. Media coverage of earnings announcements: how newsworthiness shapes media volume and tone. J Manag. 2023;49(4):1213–45. https://doi.org/10.1177/01492063221080125.
23. Barberá P, Jost JT, Nagler J, Tucker JA, Bonneau R. Tweeting from left to right: is online political communication more than an echo chamber? Psychol Sci. 2015;26(10):1531–42. https://doi.org/10.1177/0956797615594620.
24. Qin H, Li Z, Yang J. The impact of online media big data on firm performance: based on grey relation entropy method. Math Probl Eng. 2020;2020:1–7. https://doi.org/10.1155/2020/1847194.
25. Pasupuleti A, Ayyagari LR, Akuthota SR. Assessing the impact of ESG scores on market performance in polluting companies: a post-COVID-19 analysis. Discov Sustain. 2024;5(1):126. https://doi.org/10.1007/s43621-024-00338-8.
26. Zhang Y, Zhang Y, Sun Z. The impact of carbon emission trading policy on enterprise ESG performance: evidence from China. Sustainability. 2023;15(10):8279. https://doi.org/10.3390/su15108279.
27. He X, Jing Q, Chen H. The impact of environmental tax laws on heavy-polluting enterprise ESG performance: a stakeholder behavior perspective. J Environ Manage. 2023;344: 118578. https://doi.org/10.1016/j.jenvman.2023.118578.
28. Keeley AR, Chapman AJ, Yoshida K, Xie J, Imbulana J, Takeda S, Managi S. ESG metrics and social equity: investigating commensurability. Front Sustain. 2022;3: 920955. https://doi.org/10.3389/frsus.2022.920955.
29. Huang C-C, Chan Y-K, Hsieh MY. The determinants of ESG for community LOHASism sustainable development strategy. Sustainability. 2022;14(18):11429. https://doi.org/10.3390/su141811429.
30. Murphy D, McGrath D. ESG reporting—class actions, deterrence, and avoidance. Sustain Account Manage Policy J. 2013;4(2):216–35. https://doi.org/10.1108/SAMPJ-Apr-2012-0016.
31. Armstrong A. Ethics and ESG. Australasian Account Bus Finance J. 2020;14(3):6–17. https://doi.org/10.14453/aabfj.v14i3.2.
32. Becker MG, Martin F, Walter A. The power of ESG transparency: the effect of the new SFDR sustainability labels on mutual funds and individual investors. Financ Res Lett. 2022;47: 102708. https://doi.org/10.1016/j.frl.2022.102708.
33. Landi G, Sciarelli M. Towards a more ethical market: the impact of ESG rating on corporate financial performance. Soc Responsib J. 2019;15(1):11–27. https://doi.org/10.1108/SRJ-11-2017-0254.
34. Cucari N, Esposito De Falco S, Orlando B. Diversity of board of directors and environmental social governance: evidence from Italian listed companies: diversity of BoD and ESG: evidence from Italian listed companies. Corp Soc Responsib Environ Manag. 2018;25(3):250–66. https://doi.org/10.1002/csr.1452.
35. Yu EP, Guo CQ, Luu BV. Environmental, social and governance transparency and firm value. Bus Strateg Environ. 2018;27(7):987–1004. https://doi.org/10.1002/bse.2047.
36. Ge C. Negative media coverage and corporate ESG performance: evidence from China. Businesses. 2024;4(1):96–114. https://doi.org/10.3390/businesses4010007.
37. Bissoondoyal-Bheenick E, Brooks R, Do HX. ESG and firm performance: the role of size and media channels. Econ Model. 2023;121: 106203. https://doi.org/10.1016/j.econmod.2023.106203.
38. He F, Guo X, Yue P. Media coverage and corporate ESG performance: evidence from China. Int Rev Financ Anal. 2024;91: 103003. https://doi.org/10.1016/j.irfa.2023.103003.
39. Zou X, Jiang J, Zhang H, He H. ESG performance, media coverage and brand value. Asia Pac J Mark Logist. 2024. https://doi.org/10.1108/APJML-04-2024-0441.
40. Du Y, Li Z, Du J, Li N, Yan B. Public environmental appeal and innovation of heavy-polluting enterprises. J Clean Prod. 2019;222:1009–22. https://doi.org/10.1016/j.jclepro.2019.03.035.
41. Gulzar R, Bhat AA, Mir AA, Athari SA, Al-Adwan AS. Green banking practices and environmental performance: navigating sustainability in banks. Environ Sci Pollut Res. 2024;31(15):23211–26. https://doi.org/10.1007/s11356-024-32418-7.
42. Gao X, Xu W, Li D, Xing L. Media coverage and investment efficiency. J Empir Financ. 2021;63:270–93. https://doi.org/10.1016/j.jempfin.2021.07.002.
43. Wang J, Ye K. Media coverage and firm valuation: evidence from China. J Bus Ethics. 2015;127(3):501–11. https://doi.org/10.1007/s10551-014-2055-5.
44. Paananen M, Runesson E, Samani N. Time to clean up environmental liabilities reporting: disclosures, media exposure and market implications. Account Forum. 2021;45(1):85–116. https://doi.org/10.1080/01559982.2021.1872909.
45. Zhang Z, Chen H. Media coverage and impression management in corporate social responsibility reports: evidence from China. Sustain Account Manag Policy J. 2019;11(5):863–86. https://doi.org/10.1108/SAMPJ-10-2018-0293.
46. Fan L, Yang K, Liu L. New media environment, environmental information disclosure and firm valuation: evidence from high-polluting enterprises in China. J Clean Prod. 2020;277: 123253. https://doi.org/10.1016/j.jclepro.2020.123253.
47. Chen S. The technical impacts of the carbon tax in China. Discov Sustain. 2021;2(1):51. https://doi.org/10.1007/s43621-021-00060-9.
48. Koteyko N. Managing carbon emissions: a discursive presentation of 'market-driven sustainability' in the British media. Lang Commun. 2012;32(1):24–35. https://doi.org/10.1016/j.langcom.2011.11.001.
49. Xiong G, Luo Y. Smog, media attention, and corporate social responsibility—empirical evidence from Chinese polluting listed companies. Environ Sci Pollut Res. 2021;28(34):46116–29. https://doi.org/10.1007/s11356-020-11978-4.

50. Boykoff MT. We speak for the trees: media reporting on the environment. Annu Rev Environ Resour. 2009;34(1):431–57. https://doi.org/10.1146/annurev.environ.051308.084254.
51. Chih H-H, Chih H-L. Doing good with or without being known? Media coverage of corporate social performance and its impact on corporate financial performance. Manag Financ. 2014;40(9):883–902. https://doi.org/10.1108/MF-10-2013-0305.
52. Zhang C, Wu X. Analyst coverage and corporate ESG performance. Sustainability. 2023;15(17):12763. https://doi.org/10.3390/su151712763.
53. Zhao X, Fang L, Zhang K. Online search attention, firms' ESG and operating performance. Int Rev Econ Financ. 2023;88:223–36. https://doi.org/10.1016/j.iref.2023.06.025.
54. Capelle-Blancard G, Petit A. Every Little Helps? ESG news and stock market reaction. J Bus Ethics. 2019;157(2):543–65. https://doi.org/10.1007/s10551-017-3667-3.
55. Shipilov AV, Greve HR, Rowley TJ. Is all publicity good publicity? The impact of direct and indirect media pressure on the adoption of governance practices. Strateg Manag J. 2019;40(9):1368–93. https://doi.org/10.1002/smj.3030.
56. Fu L. Why bad news can be good news: the signaling feedback effect of negative media coverage of corporate irresponsibility. Organ Environ. 2023;36(1):98–125. https://doi.org/10.1177/10860266221108704.
57. Jia M, Tong L, Viswanath PV, Zhang Z. Word power: the impact of negative media coverage on disciplining corporate pollution. J Bus Ethics. 2016;138(3):437–58. https://doi.org/10.1007/s10551-015-2596-2.
58. Pan X, Chen X, Yang X. Examining the relationship between negative media coverage and corporate social responsibility. Bus Ethics Environ Responsib. 2022;31(3):620–33. https://doi.org/10.1111/beer.12430.
59. Cade NL. Corporate social media: how two-way disclosure channels influence investors. Acc Organ Soc. 2018;68–69:63–79. https://doi.org/10.1016/j.aos.2018.03.004.
60. Bhat AA, Mir AA, Allie AH, Ahmad Lone M, Al-Adwan AS, Jamali D, Riyaz I. Unlocking corporate social responsibility and environmental performance: mediating role of green strategy, innovation, and leadership. Innov Green Dev. 2024;3(2): 100112. https://doi.org/10.1016/j.igd.2023.100112.
61. Chen Y-S, Lai S-B, Wen C-T. The influence of green innovation performance on corporate advantage in Taiwan. J Bus Ethics. 2006;67(4):331–9. https://doi.org/10.1007/s10551-006-9025-5.
62. Wüstenhagen R, Wolsink M, Bürer MJ. Social acceptance of renewable energy innovation: an introduction to the concept. Energy Policy. 2007;35(5):2683–91. https://doi.org/10.1016/j.enpol.2006.12.001.
63. Sun H, Edziah BK, Sun C, Kporsu AK. Institutional quality, green innovation and energy efficiency. Energy Policy. 2019;135: 111002. https://doi.org/10.1016/j.enpol.2019.111002.
64. Sun Y, Gao P, Tian W, Guan W. Green innovation for resource efficiency and sustainability: empirical analysis and policy. Resour Policy. 2023;81: 103369. https://doi.org/10.1016/j.resourpol.2023.103369.
65. Brunel C. Green innovation and green Imports: links between environmental policies, innovation, and production. J Environ Manage. 2019;248: 109290. https://doi.org/10.1016/j.jenvman.2019.109290.
66. Dangelico RM, Pujari D, Pontrandolfo P. Green product innovation in manufacturing firms: a sustainability-oriented dynamic capability perspective. Bus Strateg Environ. 2017;26(4):490–506. https://doi.org/10.1002/bse.1932.
67. Shah KJ, Pan S-Y, Lee I, Kim H, You Z, Zheng J-M, Chiang P-C. Green transportation for sustainability: review of current barriers, strategies, and innovative technologies. J Clean Prod. 2021;326: 129392. https://doi.org/10.1016/j.jclepro.2021.129392.
68. Jia L, Xu R, Shen ZY, Song M. Which type of innovation is more conducive to inclusive green growth: independent innovation or imitation innovation? J Clean Prod. 2023;406: 137026. https://doi.org/10.1016/j.jclepro.2023.137026.
69. Razzaq A, Sharif A, Ozturk I, Skare M. Inclusive infrastructure development, green innovation, and sustainable resource management: evidence from China's trade-adjusted material footprints. Resour Policy. 2022;79: 103076. https://doi.org/10.1016/j.resourpol.2022.103076.
70. Feng G-F, Zheng M. Economic policy uncertainty and renewable energy innovation: international evidence. Innov Green Dev. 2022;1(2): 100010. https://doi.org/10.1016/j.igd.2022.100010.
71. Hu J, Wang K-H, Su CW, Umar M. Oil price, green innovation and institutional pressure: a China's perspective. Resour Policy. 2022;78: 102788. https://doi.org/10.1016/j.resourpol.2022.102788.
72. Sun Y, Razzaq A. Composite fiscal decentralisation and green innovation: imperative strategy for institutional reforms and sustainable development in OECD countries. Sustain Dev. 2022;30(5):944–57. https://doi.org/10.1002/sd.2292.
73. Huang X, Liu W, Zhang Z, Zou X, Li P. Quantity or quality: environmental legislation and corporate green innovations. Ecol Econ. 2023;204: 107684. https://doi.org/10.1016/j.ecolecon.2022.107684.
74. Huang Z, Liao G, Li Z. Loaning scale and government subsidy for promoting green innovation. Technol Forecast Soc Chang. 2019;144:148–56. https://doi.org/10.1016/j.techfore.2019.04.023.
75. Liu J, Zhao M, Wang Y. Impacts of government subsidies and environmental regulations on green process innovation: a nonlinear approach. Technol Soc. 2020;63: 101417. https://doi.org/10.1016/j.techsoc.2020.101417.
76. Gürlek M, Tuna M. Reinforcing competitive advantage through green organizational culture and green innovation. Serv Ind J. 2018;38(7–8):467–91. https://doi.org/10.1080/02642069.2017.1402889.
77. He J, Lei Y, Fu X. Do consumer's green preference and the reference price effect improve green innovation? A theoretical model using the food supply chain as a case. Int J Environ Res Public Health. 2019;16(24):5007. https://doi.org/10.3390/ijerph16245007.
78. Melander L, Pazirandeh A. Collaboration beyond the supply network for green innovation: insight from 11 cases. Supply Chain Manag. 2019;24(4):509–23. https://doi.org/10.1108/SCM-08-2018-0285.
79. Chen Z, Jin J, Li M. Does media coverage influence firm green innovation? The moderating role of regional environment. Technol Soc. 2022;70: 102006. https://doi.org/10.1016/j.techsoc.2022.102006.
80. Shah MI, Foglia M, Shahzad U, Fareed Z. Green innovation, resource price and carbon emissions during the COVID-19 times: new findings from wavelet local multiple correlation analysis. Technol Forecast Soc Chang. 2022;184: 121957. https://doi.org/10.1016/j.techfore.2022.121957.

81. Kunapatarawong R, Martínez-Ros E. Towards green growth: how does green innovation affect employment? Res Policy. 2016;45(6):1218–32. https://doi.org/10.1016/j.respol.2016.03.013.
82. Asni N, Agustia D. Does corporate governance induce green innovation? An emerging market evidence. Corp Gov Int J Bus Soc. 2022;22(7):1375–89. https://doi.org/10.1108/CG-10-2021-0389.
83. Khan U, Liu W. The link between green innovations, corporate performance, ESG activities, and sharing economy. Environ Sci Pollut Res. 2023;30(32):78763–75. https://doi.org/10.1007/s11356-023-27722-7.
84. Zheng J, Khurram MU, Chen L. Can green innovation affect ESG ratings and financial performance? Evidence from Chinese GEM listed companies. Sustainability. 2022;14(14):14. https://doi.org/10.3390/su14148677.
85. Chen S, Golley J. 'Green' productivity growth in China's industrial economy. Energy Econ. 2014;44:89–98. https://doi.org/10.1016/j.eneco.2014.04.002.
86. Lei S, Zhang Y. The role of the media in socially responsible investing. Int J Bank Mark. 2020;38(4):823–41. https://doi.org/10.1108/IJBM-09-2019-0332.
87. Hammami A, Hendijani Zadeh M. Audit quality, media coverage, environmental, social, and governance disclosure and firm investment efficiency: evidence from Canada. Int J Account Inf Manag. 2019;28(1):45–72. https://doi.org/10.1108/IJAIM-03-2019-0041.
88. Lyytimäki J, Nygrén NA, Pulkka A, Rantala S. Energy transition looming behind the headlines? Newspaper coverage of biogas production in Finland. Energy Sustain Soc. 2018;8(1):15. https://doi.org/10.1186/s13705-018-0158-z.
89. Rochyadi-Reetz M, Arlt D, Wolling J, Bräuer M. Explaining the media's framing of renewable energies: an international comparison. Front Environ Sci. 2019;7:119. https://doi.org/10.3389/fenvs.2019.00119.
90. Ren C, Ting IWK, Lu W, Kweh QL. Nonlinear effects of ESG on ENERGY-ADJUSTED firm efficiency: evidence from the stakeholder engagement of apple incorporated. Corp Soc Responsib Environ Manag. 2022;29(5):1231–46. https://doi.org/10.1002/csr.2266.
91. Bendickson JS, Irwin JG, Cowden BJ, McDowell WC. Entrepreneurial ecosystem knowledge spillover in the face of institutional voids: groups, issues, and actions. Knowl Manag Res Pract. 2021;19(1):117–26. https://doi.org/10.1080/14778238.2020.1768810.
92. Sun X, Lin B, Gao J, Lin Y. Media coverage, tourism growth, and spatial spillover: evidence from China. Tour Anal. 2018;23(4):503–15. https://doi.org/10.3727/108354218X15391984820459.
93. McCluskey JJ, Kalaitzandonakes N, Swinnen J. Media coverage, public perceptions, and consumer behavior: insights from new food technologies. Annu Rev Resour Econ. 2016;8(1):467–86. https://doi.org/10.1146/annurev-resource-100913-012630.
94. Kelton AS, Pennington RR. Internet financial reporting: the effects of information presentation format and content differences on investor decision making. Comput Hum Behav. 2012;28(4):1178–85. https://doi.org/10.1016/j.chb.2012.01.028.
95. Long H, Feng G-F, Chang C-P. How does ESG performance promote corporate green innovation? Econ Chang Restruct. 2023;56(4):2889–913. https://doi.org/10.1007/s10644-023-09536-2.
96. Zhang D, Meng L, Zhang J. Environmental subsidy disruption, skill premiums and ESG performance. Int Rev Financ Anal. 2023;90: 102862. https://doi.org/10.1016/j.irfa.2023.102862.
97. Luo Y, Nyarko Mensah C, Lu Z, Wu C. Environmental regulation and green total factor productivity in China: a perspective of Porter's and Compliance Hypothesis. Ecol Ind. 2022;145: 109744. https://doi.org/10.1016/j.ecolind.2022.109744.
98. Huang Y, Li X, Liu Y. The impact of environmental regulation or bargaining power on green total factor productivity: evidence from Taiwan-funded enterprises in Chinese mainland. Front Environ Sci. 2022;10: 982430. https://doi.org/10.3389/fenvs.2022.982430.
99. Yao S, Pan Y, Sensoy A, Uddin GS, Cheng F. Green credit policy and firm performance: what we learn from China. Energy Econ. 2021;101: 105415. https://doi.org/10.1016/j.eneco.2021.105415.
100. Rudkin B, Kimani D, Ullah S, Ahmed R, Farooq SU. Hide-and-seek in corporate disclosure: evidence from negative corporate incidents. Corp Gov Int J Bus Soc. 2019;19(1):158–75. https://doi.org/10.1108/CG-05-2018-0164.
101. Asimakopoulos P, Asimakopoulos S, Li X. The role of environmental, social, and governance rating on corporate debt structure. J Corp Finan. 2023;83: 102488. https://doi.org/10.1016/j.jcorpfin.2023.102488.
102. Yang Y, Han J. Digital transformation, financing constraints, and corporate environmental, social, and governance performance. Corp Soc Responsib Environ Manag. 2023. https://doi.org/10.1002/csr.2546.
103. Meng-tao C, Da-peng Y, Wei-qi Z, Qi-jun W. How does ESG disclosure improve stock liquidity for enterprises—empirical evidence from China. Environ Impact Assess Rev. 2023;98: 106926. https://doi.org/10.1016/j.eiar.2022.106926.
104. Husted BW, Sousa-Filho JMD. Board structure and environmental, social, and governance disclosure in Latin America. J Bus Res. 2019;102:220–7. https://doi.org/10.1016/j.jbusres.2018.01.017.
105. Mbalyohere C, Lawton T, Boojihawon R, Viney H. Corporate political activity and location-based advantage: MNE responses to institutional transformation in Uganda's electricity industry. J World Bus. 2017;52(6):743–59. https://doi.org/10.1016/j.jwb.2017.06.006.
106. Pantzalis C. Does location matter? An empirical analysis of geographic scope and MNC market valuation. J Int Bus Stud. 2001;32(1):133–55. https://doi.org/10.1057/palgrave.jibs.8490942.
107. Du P, Huang S, Hong Y, Wu W. Can FinTech improve corporate environmental, social, and governance performance?—A study based on the dual path of internal financing constraints and external fiscal incentives. Front Environ Sci. 2022. https://doi.org/10.3389/fenvs.2022.1061454.
108. Wang L, Le Q, Peng M, Zeng H, Kong L. Does central environmental protection inspection improve corporate environmental, social, and governance performance? Evidence from China. Bus Strateg Environ. 2023;32(6):2962–84. https://doi.org/10.1002/bse.3280.
109. Matsumoto A, Merlone U, Szidarovszky F. Some notes on applying the Herfindahl-Hirschman index. Appl Econ Lett. 2012;19(2):181–4. https://doi.org/10.1080/13504851.2011.570705.
110. Kim J, Kim M, Im S, Choi D. Competitiveness of E commerce firms through ESG logistics. Sustainability. 2021;13(20):11548. https://doi.org/10.3390/su132011548.
111. Jeong HJ, Chung DS. The story of corporate social responsibility: an attribution analysis of the coverage of U.S. corporate responsibility cases. J Media Ethics. 2022;37(4):266–80. https://doi.org/10.1080/23736992.2022.2142126.
112. Chai S, Cao M, Li Q, Ji Q, Liu Z. Exploring the nexus between ESG disclosure and corporate sustainable growth: Moderating role of media attention. Financ Res Lett. 2023;58: 104519. https://doi.org/10.1016/j.frl.2023.104519.

113. Cui W, Chen X, Xia W, Hu Y. Influence of media attention on the quality of environmental, social, and governance information disclosure in enterprises: an adjustment effect based on the shareholder relationship network. Sustainability. 2023;15(18):13919. https://doi.org/10.3390/su151813919.
114. Tan Y, Zhu Z. The effect of ESG rating events on corporate green innovation in China: the mediating role of financial constraints and managers' environmental awareness. Technol Soc. 2022;68: 101906. https://doi.org/10.1016/j.techsoc.2022.101906.
115. Deng X, Li W, Ren X. More sustainable, more productive: evidence from ESG ratings and total factor productivity among listed Chinese firms. Financ Res Lett. 2023;51: 103439. https://doi.org/10.1016/j.frl.2022.103439.
116. Zheng M, Feng G, Jiang R, Chang C. Does environmental, social, and governance performance move together with corporate green innovation in China? Bus Strategy Environ. 2022. https://doi.org/10.1002/bse.3211.
117. Driscoll JC, Kraay AC. Consistent covariance matrix estimation with spatially dependent panel data. Rev Econ Stat. 1998;80(4):549–60. https://doi.org/10.1162/003465398557825.